

Project 15: Moving Average Crossover Strategy

Implementation and Analysis

Objective: Build a complete moving average crossover trading strategy with proper backtesting, visualization, and performance evaluation.

Requirements:

1. Select at least 3 different stocks from various sectors and download at least 3 years of historical data.
2. Implement a moving average crossover strategy with the following components:
 - Calculate both Simple Moving Averages (SMA) and Exponential Moving Averages (EMA)
 - Test at least 3 different pairs of moving average periods (e.g., 10/30, 20/50, 50/200)
 - Generate clear entry and exit signals based on the crossovers
3. Create a comprehensive backtesting framework that:
 - Tracks position status (in market vs. out of market)
 - Calculates daily returns for both the strategy and buy-and-hold
 - Handles position sizing (start with fixed positions, then implement percentage-based sizing)
 - Simulates realistic trading by accounting for basic transaction costs
4. Develop detailed visualizations including:
 - Price charts with both moving averages plotted
 - Buy and sell signals marked on the chart